LECTURE#07

MULTIVARIABLE CALCULUS

CHAIN RULE IN FUNCTION OF ONE VARIABLE

Given that $w = f(x)$ and $\mathrm{x} = g(t)$ we find $\frac{dw}{dt}$ as follows

From $w = f(x)$, we get $\frac{dw}{dx}$

From $\mathrm{x} = g(t)$, we get $\frac{dx}{dt}$

Then $$\frac{dw}{dt} = \frac{dw}{dx} . \frac{dx}{dt}$$

EXAMPLE 01

$$\boldsymbol{Find\ \frac{dw}{dt},\quad if\quad w = x + 4,\quad x = sint.}$$

Solution: By substitution $w = sint + 4$

$$\frac{dw}{dt} = cost$$

Now by chain rule, $w = x + 4,\ x = sint$

$$\frac{dw}{dx} = 1,\qquad \frac{dx}{dt} = cost$$

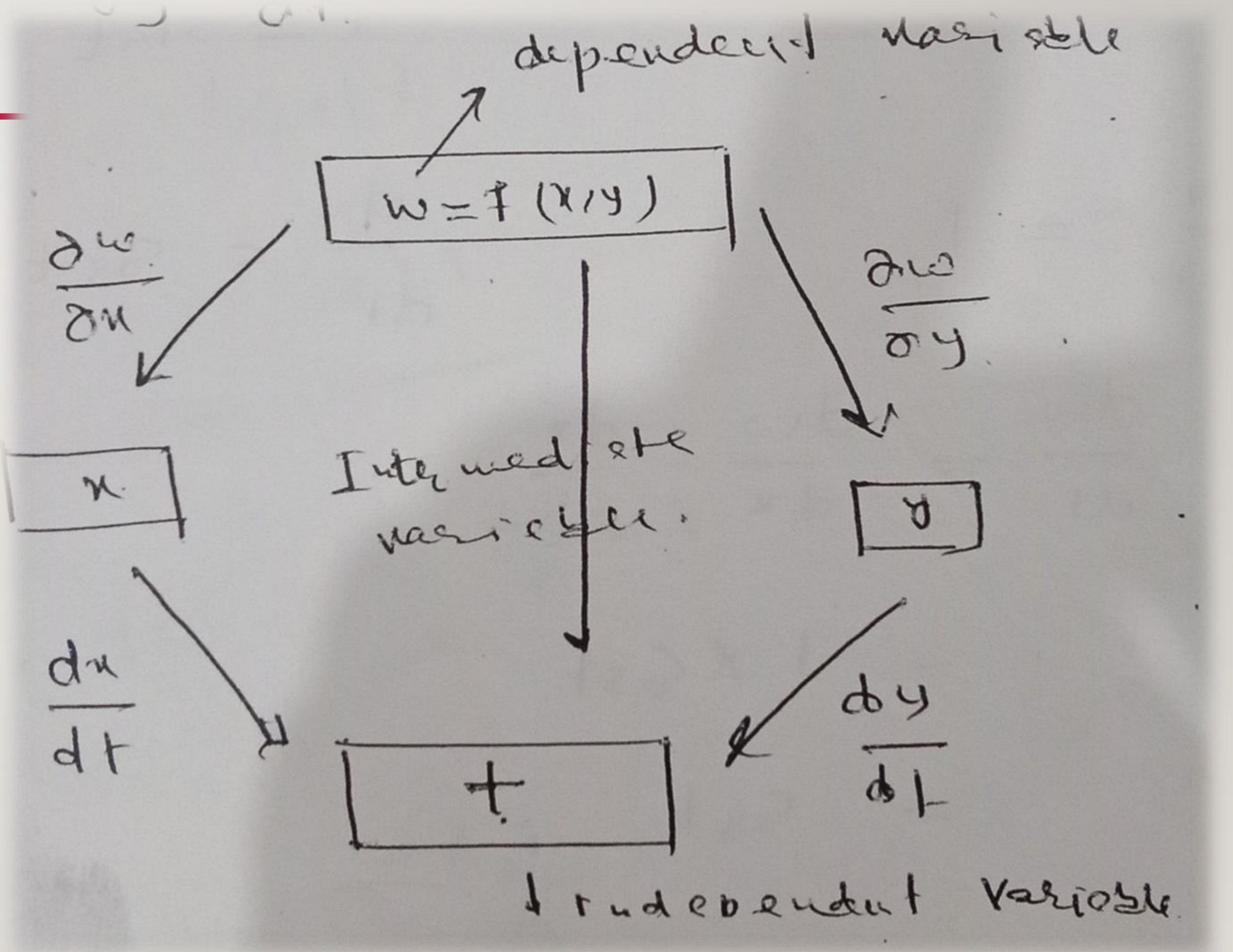
$$\frac{dw}{dt} = \frac{dw}{dx}.\frac{dx}{dt} = 1.cost = cost$$

CHAIN RULE IN MULTIVARIABLE FUNCTIONS

For two variables:

$$w = f(x, y), x = g(t), y = f(t)$$

$$\frac{dw}{dt} = \frac{\partial w}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial w}{\partial y} \cdot \frac{dy}{dt}$$



EXAMPLE 01

$$\boldsymbol{Find \quad \frac{dw}{dt}, if \quad w = xy, \qquad x = cost, \qquad y = sint.}$$

Solution: By substitution $w = (cost)(sint) = \frac{1}{2}.2sintcost$

$$= \frac{1}{2}sin2t$$

$$\frac{dw}{dt} = \frac{1}{2}cos2t.2 = cos2t$$

By chain rule

$$\frac{\partial w}{\partial y} = x, \qquad \frac{\partial w}{\partial x} = y$$

$$\frac{dx}{dt} = -sint, \qquad \frac{dy}{dt} = cost$$

$$\begin{aligned}\frac{dw}{dt} &= \frac{\partial w}{\partial x}.\frac{dx}{dt} + \frac{\partial w}{\partial y}.\frac{dy}{dt} \\ &= y(-sint) + x(cost) \\ &= (sint)(-sint) + (cost)(cost) \\ &= -sin^2t + cos^2t \\ &= cos2t\end{aligned}$$

EXAMPLE 02

$$\boldsymbol{Find\ \ \frac{dz}{dt}, if\ \ \ z = 3x^2y^3, \qquad x = t^4, \qquad y = t^2.}$$

Solution:

$$\frac{\partial z}{\partial x} = 6xy^3, \qquad \frac{\partial z}{\partial y} = 9x^2y^2$$

$$\frac{dx}{dt} = 4t^3, \qquad \frac{dy}{dt} = 2t$$

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} . \frac{dx}{dt} + \frac{\partial z}{\partial y} . \frac{dy}{dt}$$

$$= (6xy^3)(4t^3) + (9x^2y^2)(2t)$$

By putting values of x and y, we have

$$= 6t^4(t^2)^3(4t^3) + 9(t^4)^2(t^2)^2(2t)$$
$$= 6(t^4)(t^6)(4t^3) + 9(t^8)(t^4)\,(2t)$$
$$= 24t^{13} + 18t^{13}$$
$$\frac{dz}{dt} = 42t^{13}$$

Example 03

$$\boldsymbol{z = \sqrt{1 + x - 2xy^4}, \qquad x = ln(t), \qquad y = t}$$

Solution: $\frac{\partial z}{\partial x} = \frac{1}{2\sqrt{1+x-2xy^4}}(1 - 2y^4),$ $\qquad \frac{\partial z}{\partial y} = \frac{1}{2\sqrt{1+x-2xy^4}}(-8xy^3)$

$$\frac{\partial z}{\partial x} = \frac{(1 - 2y^4)}{2\sqrt{1 + x - 2xy^4}}, \qquad \frac{\partial z}{\partial y} = \frac{-4xy^3}{\sqrt{1 + x - 2xy^4}}$$

$$\frac{dx}{dt} = \frac{1}{t}, \qquad \frac{dy}{dt} = 1$$

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial z}{\partial y} \cdot \frac{dy}{dt}$$

$$= \left(\frac{(1-2y^4)}{2\sqrt{1+x-2xy^4}}\right)\left(\frac{1}{t}\right) + \left(\frac{(-4xy^3)}{\sqrt{1+x-2xy^4}}\right)(1)$$

$$= \left(\frac{1}{\sqrt{1+x-2xy^4}}\right)\left(\frac{1-2y^4}{2t} - 4xy^3\right)$$

$$= \left(\frac{1}{\sqrt{1+lnt-2(lnt)t^4}}\right)\left(\frac{1-2t^4}{2t} - 4(lnt)t^3\right)$$

$$= \left(\frac{1}{\sqrt{1+lnt-2t^4 lnt}}\right)\left(\frac{1}{2t} - t^3 - 4t^3 lnt\right)$$

EXAMPLE 04

$$\boldsymbol{z = ln(2x^2 + y), \qquad x = \sqrt{t}, \qquad y = t^{2/3}}$$

Solution:

$$\frac{\partial z}{\partial x} = \frac{1}{2x^2 + y}(4x), \qquad \frac{\partial z}{\partial y} = \frac{1}{2x^2 + y}(1)$$

$$\frac{\partial z}{\partial x} = \frac{4x}{2x^2 + y}, \qquad \frac{\partial z}{\partial y} = \frac{1}{2x^2 + y}$$

$$\frac{dx}{dt} = \frac{1}{2\sqrt{t}}, \qquad \frac{dy}{dt} = \frac{2}{3}t^{-1/3}$$

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial z}{\partial y} \cdot \frac{dy}{dt}$$

$$\frac{dz}{dt} = \left(\frac{4x}{2x^2 + y}\right) \cdot \left(\frac{1}{2\sqrt{t}}\right) + \left(\frac{1}{2x^2 + y}\right) \cdot \left(\frac{2}{3} t^{-1/3}\right)$$

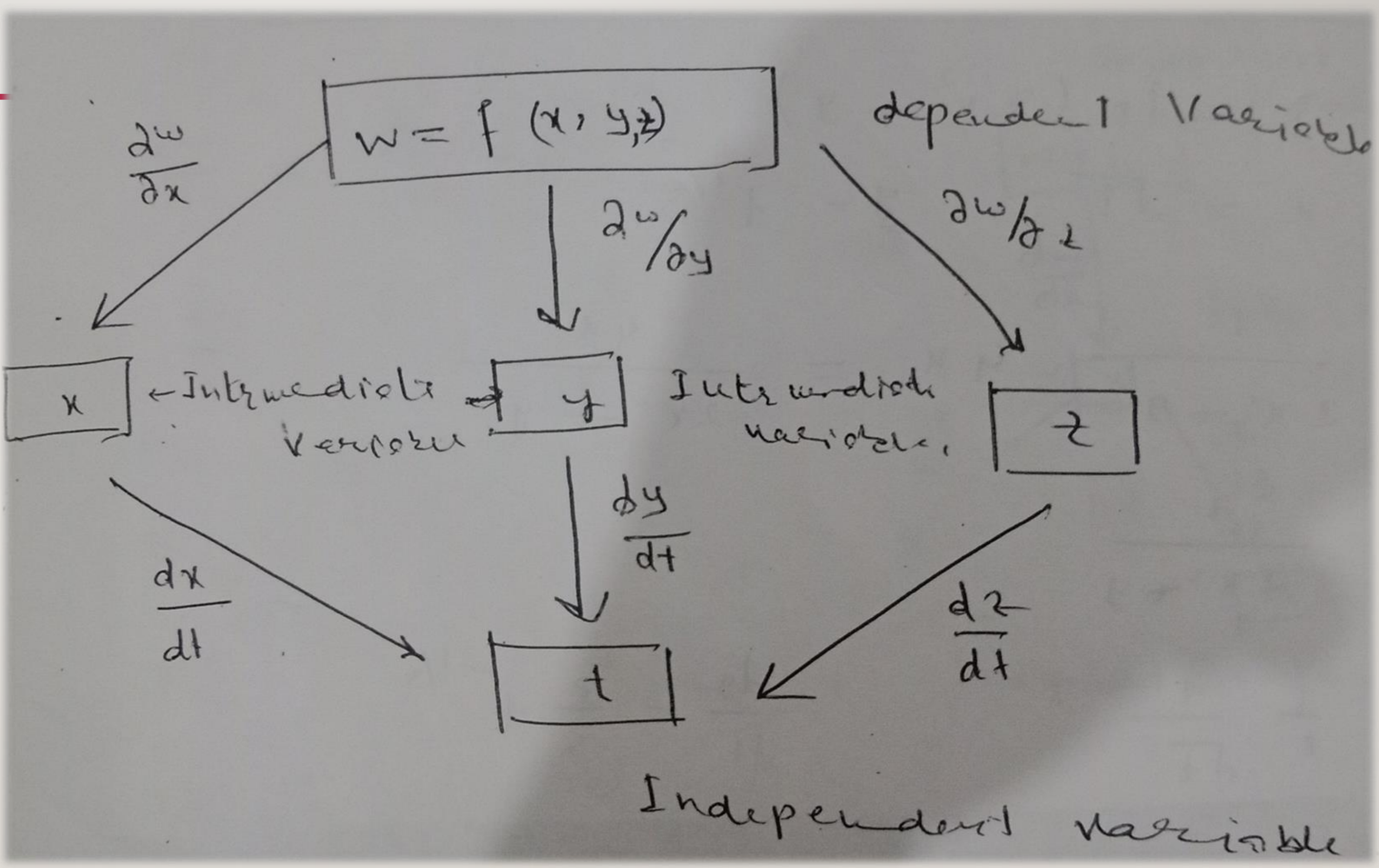
$$= \left(\frac{4(\sqrt{t})}{2(\sqrt{t})^2 + t^{2/3}}\right) \cdot \left(\frac{1}{2\sqrt{t}}\right) + \left(\frac{1}{2(\sqrt{t})^2 + t^{2/3}}\right) \cdot \left(\frac{2}{3} t^{-1/3}\right)$$

$$= \left(\frac{4(\sqrt{t})}{2t + t^{2/3}}\right) \cdot \left(\frac{1}{2\sqrt{t}}\right) + \left(\frac{2}{3\left(2t + t^{\frac{2}{3}}\right)}\right) \cdot \left(\frac{1}{t^{-\frac{1}{3}}}\right)$$

$$\frac{dz}{dt} = \frac{6t^{1/3} + 2}{3(2t + t^{2/3})t^{1/3}}$$

CONT…

$$w = f(x, y, z),$$
$$x = g(t), y = f(t), z = h(t)$$
$$\frac{dw}{dt} = \frac{\partial w}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial w}{\partial y} \cdot \frac{dy}{dt} + \frac{\partial w}{\partial z} \cdot \frac{dz}{dt}$$



EXAMPLE 01

Find $\dfrac{dw}{dt}$ $\boldsymbol{if \;\; w = x^2 + y + z + 4, \;\; x = e^t, \;\; y = cost, \;\; z = t + 4}$

Solution:

$$\frac{\partial w}{\partial x} = 2x, \qquad \frac{\partial w}{\partial y} = 1, \qquad \frac{\partial w}{\partial z} = 1$$

$$\frac{dx}{dt} = e^t, \qquad \frac{dy}{dt} = -sint, \qquad \frac{dz}{dt} = 1$$

$$\frac{dw}{dt} = \frac{\partial w}{\partial x}.\frac{dx}{dt} + \frac{\partial w}{\partial y}.\frac{dy}{dt} + \frac{\partial w}{\partial z}.\frac{dz}{dt}$$

$$= (2x).(e^t) + (1).(-sint) + (1).(1)$$

$$= 2(e^t).(e^t) - sint \; + 1$$

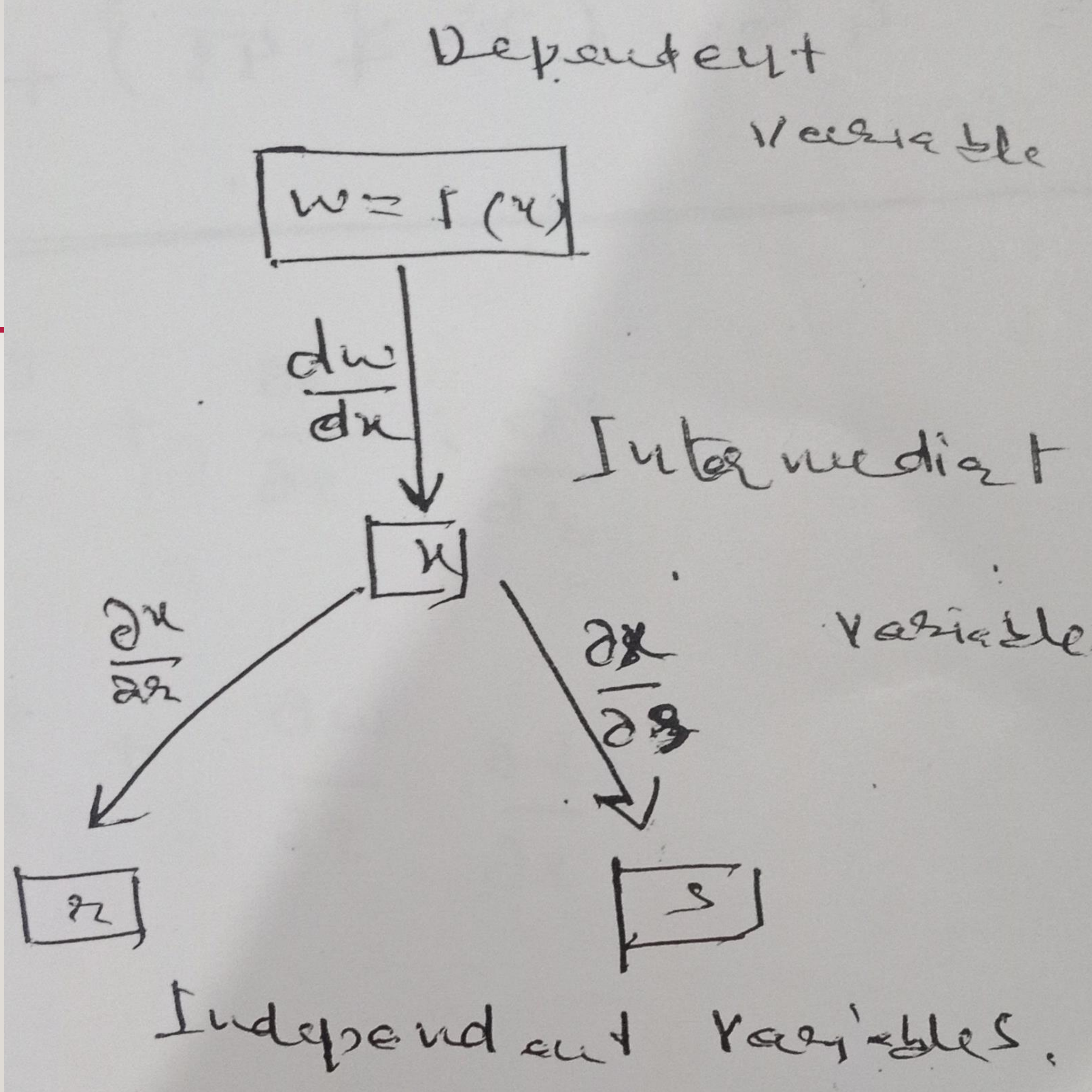
$$\frac{dw}{dt} = 2e^{2t} - sint \; + 1$$

CONT…

$$w = f(x), \qquad x = g(r, s)$$

$$\frac{\partial w}{\partial r} = \frac{dw}{dx} \cdot \frac{\partial x}{\partial r}$$

$$\frac{\partial w}{\partial s} = \frac{dw}{dx} \cdot \frac{\partial x}{\partial s}$$



EXAMPLE 01

Find $\frac{\partial w}{\partial r}, \frac{\partial w}{\partial s}$ ***if*** $\boldsymbol{w = sinx + x^2, \quad x = 3r + 4s.}$

Solution:

$$\frac{dw}{dx} = cosx + 2x,$$

$$\frac{\partial x}{\partial r} = 3, \qquad \frac{\partial x}{\partial s} = 4,$$

$$\frac{\partial w}{\partial r} = \frac{dw}{dx} \cdot \frac{\partial x}{\partial r}$$

$$\frac{\partial w}{\partial r} = (cosx + 2x).(3) = 3(cosx + 2x)$$

$$= 3\big(\cos(3r + 4s) + 2(3r + 4s)\big)$$

$$= 3\cos(3r + 4s) + 6(3r + 4s)$$

$$\frac{\partial w}{\partial r} = 3\cos(3r + 4s) + 18r + 24s$$

$$\frac{\partial w}{\partial s} = \frac{dw}{dx} . \frac{\partial x}{\partial s}$$

$$\frac{\partial w}{\partial s} = (cosx + 2x).(4)$$

$$= 4(cosx + 2x)$$

$$= 4\big(\cos(3r + 4s) + 2(3r + 4s)\big)$$

$$= 4\cos(3r + 4s) + 8(3r + 4s)$$

$$\frac{\partial w}{\partial s} = 4\cos(3r + 4s) + 24r + 32s$$